

REVERSE COMPUTATION RESEARCH HUB

GROUP 19

PROJECT OVERVIEW

Mason Harniess

Goals:

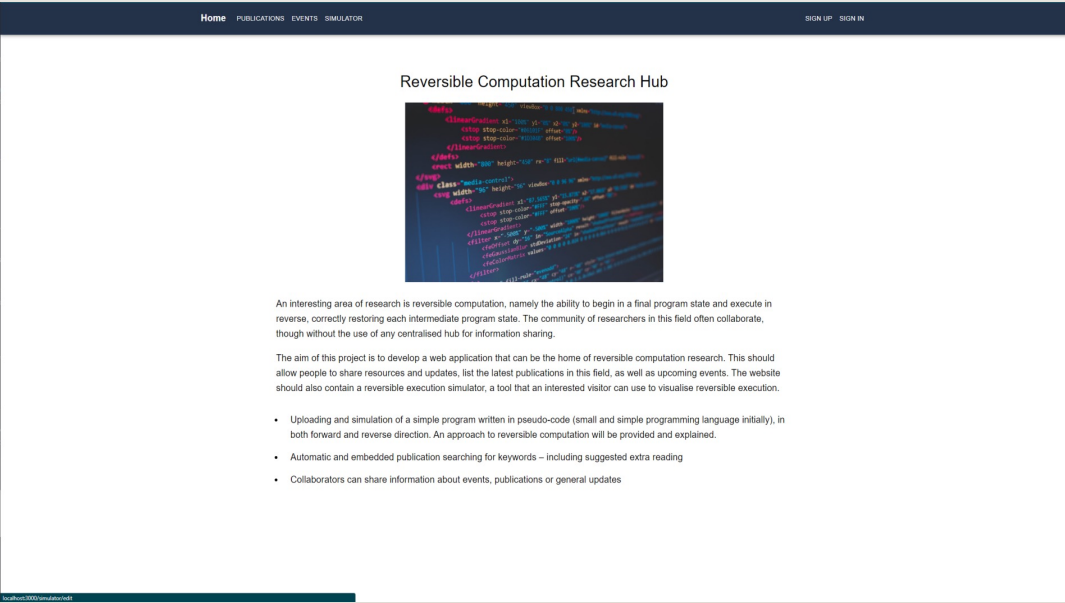
- Develop a web application to act as the home of reversible computation research.
- Allow people to share resources and updates, mainly publications and events.
- Demonstrate reverse computation via a simulator

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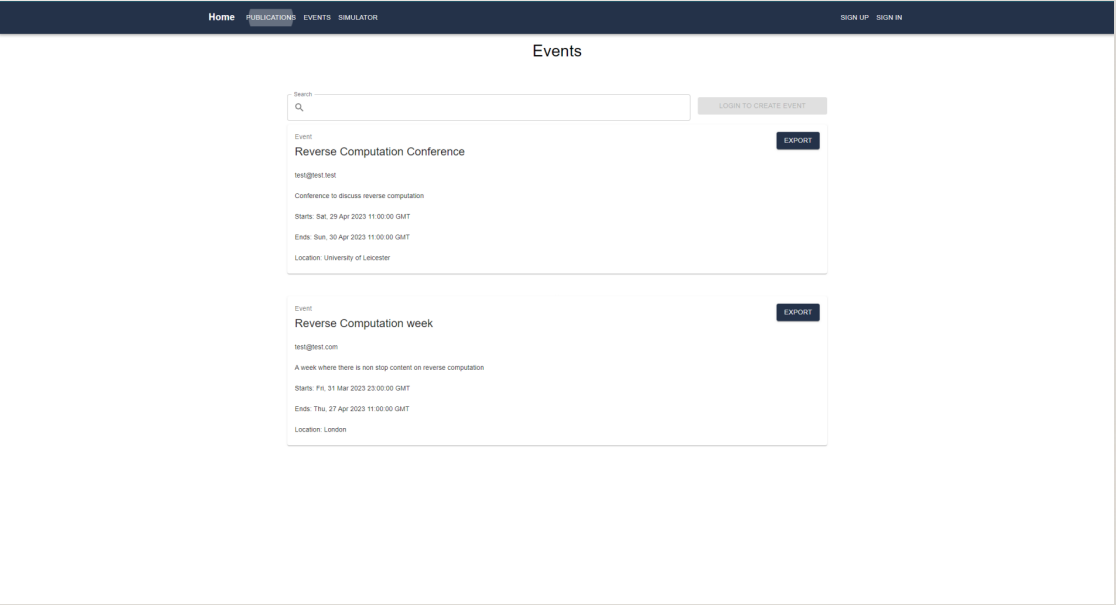
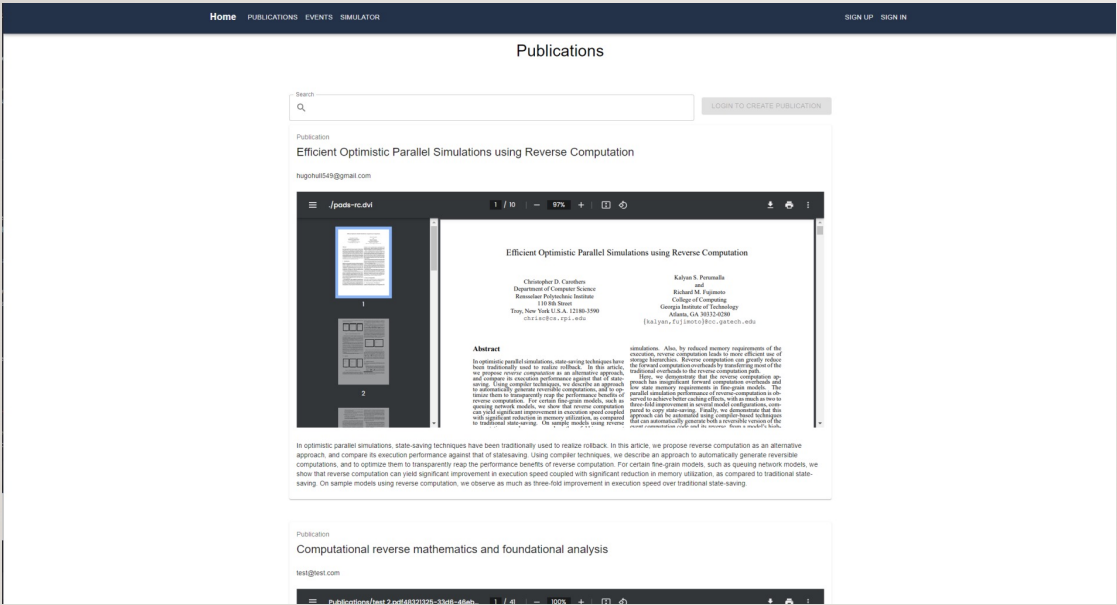


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Goals:

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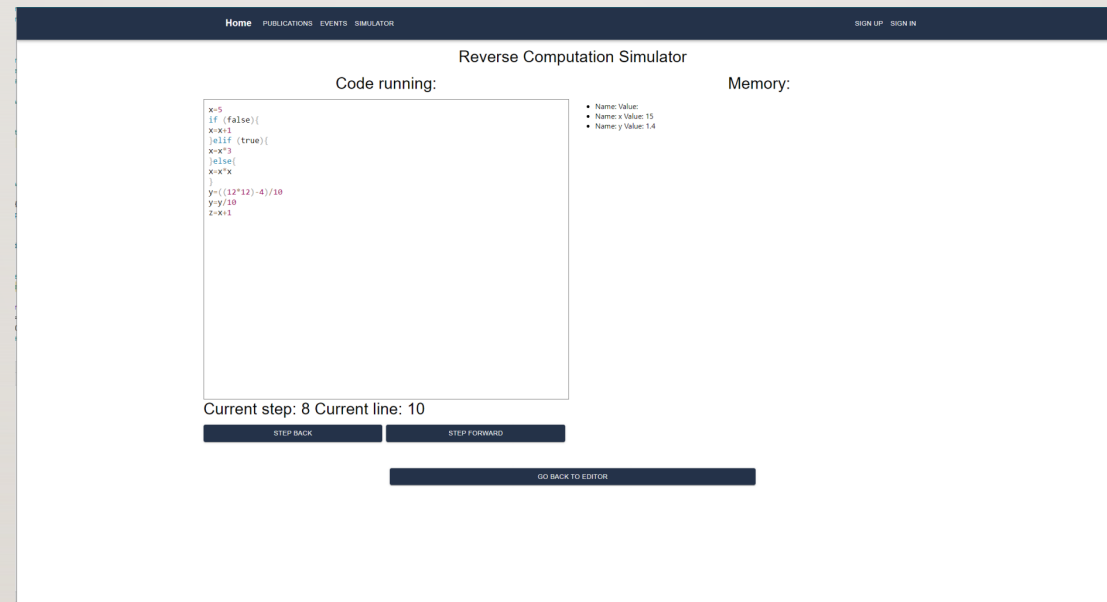


PROJECT OVERVIEW

Mason Harniess

Goals:

- Demonstrate reverse computation via a simulator



LEGAL, SOCIAL AND ETHICAL ISSUES

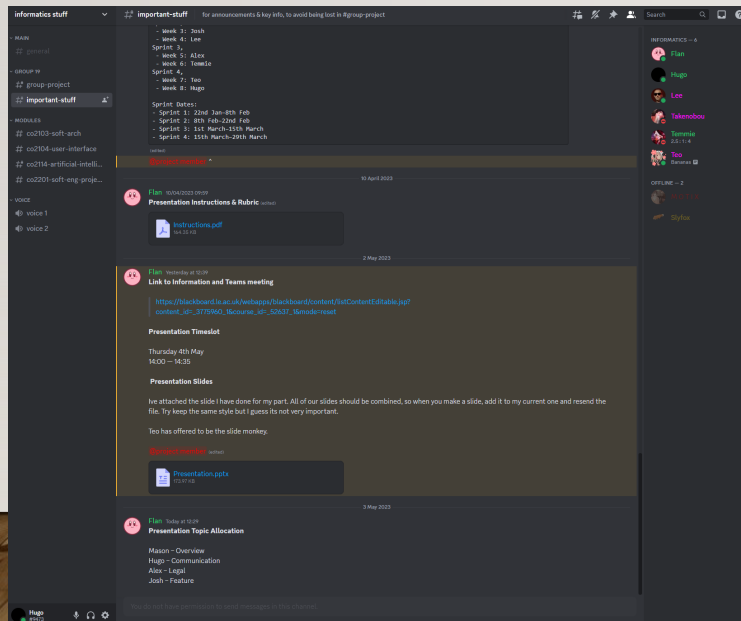
Alex Martin

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- Copyright: Respecting intellectual property rights of research materials.
 - Fair Use: Ensuring proper citation and usage of published materials.
 - Accessibility: Ensuring equal access to the web app for people with disabilities.
 - Security: Safeguarding the web app from unauthorized access or malicious attacks.
 - Misinformation: Promoting transparency and accountability to ensure reliable and accurate content sharing.

HOW WE COMMUNICATED

Hugo Hull

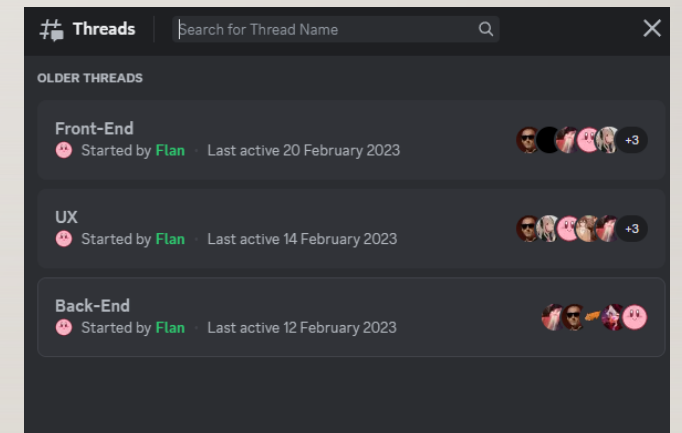
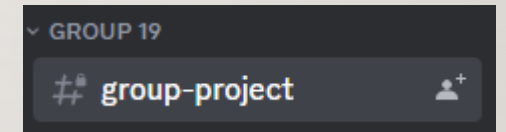
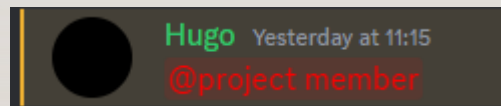
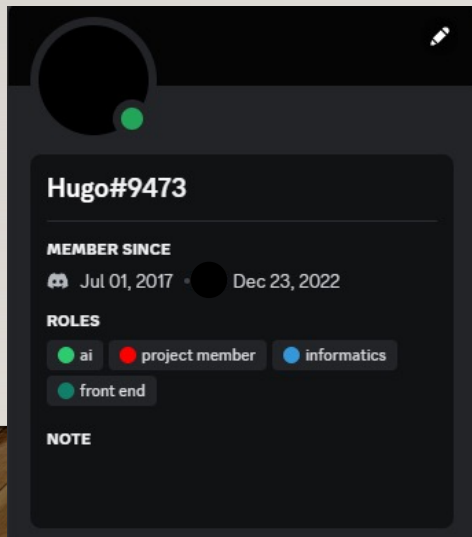
- Communication: “the imparting or exchanging of information by speaking, writing, or using some other medium.”



USING DISCORD

Hugo Hull

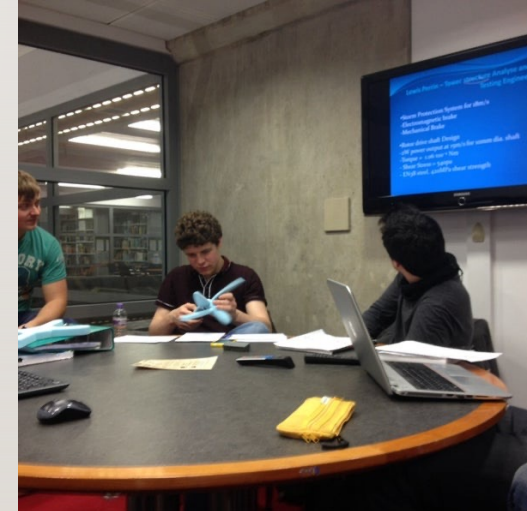
- Like everything it had its positives and negatives



GROUP MEETINGS

Hugo Hull

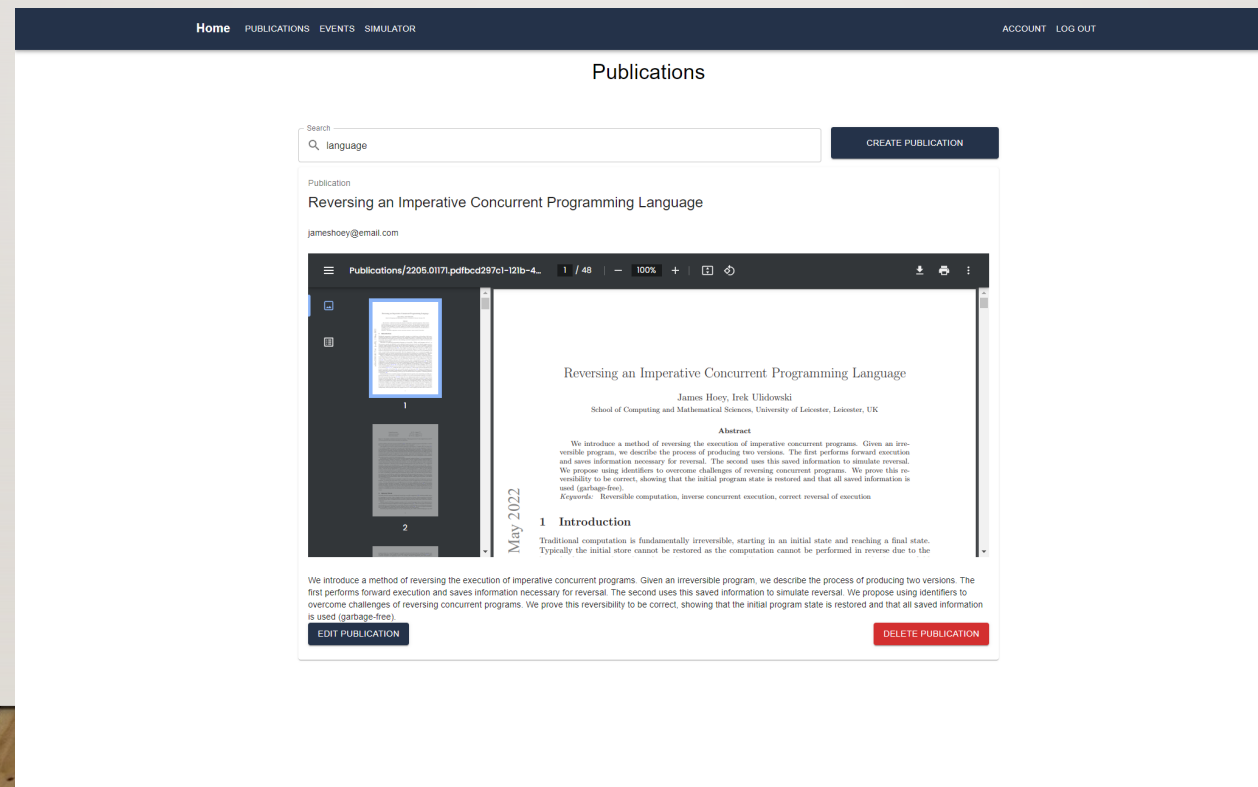
- We met in the Group Study Rooms of the David Wilson Library.



WHETHER IT WORKED WELL OR NOT

Hugo Hull

- I think we communicated very well overall.



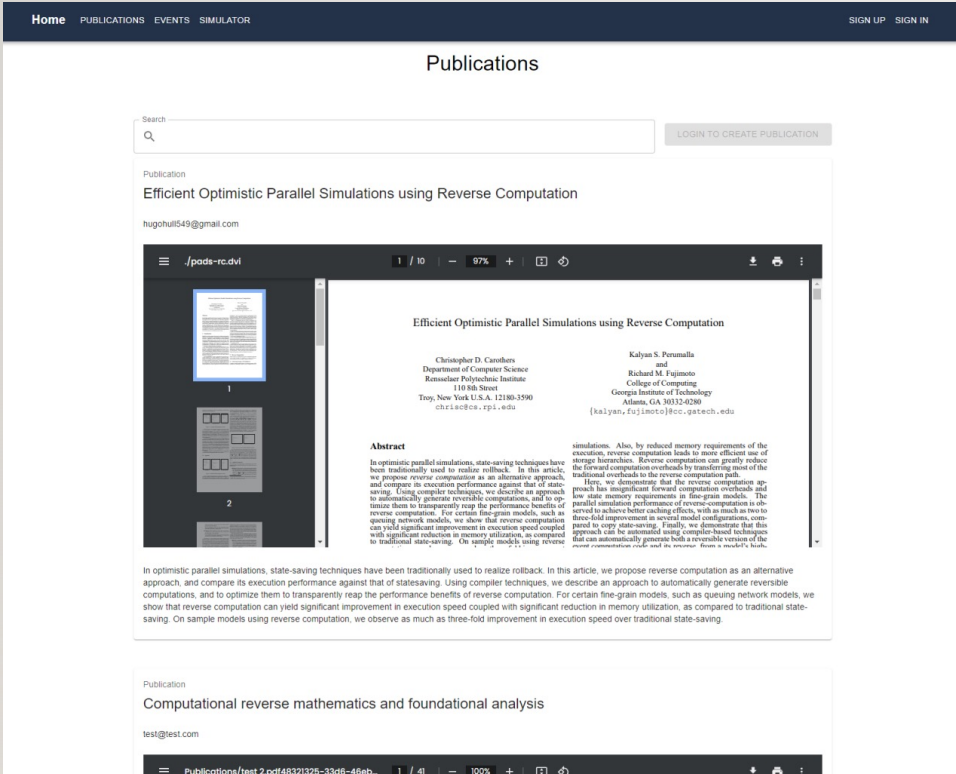
NON-TECHNICAL SKILLS AND KNOWLEDGE I LEARN DURING A PROJECT

Klaudiusz Halek

- Leadership as this helped me become more assertive and confident in myself.
- Adaptability as I had to try new things as well as polish my current skill to better degree.
- Time management because I had to balance my time between other courses.

MAIN FEATURE: PUBLICATIONS

Josh Hull



▼ — Publications

▶

4bf343dc-96d2-4585-afb3-0f78a0ca2fcd

▶

581ff08b-0624-4f73-8f5d-5c7541a68cc6

▼

5fed3116-7a73-46b4-8394-f173935cb53f

— account: "test@test.com"

— description: "Reverse mathematics studies which subsystems of second order

— filename: "test 2.pdf48321325-33d6-46eb-ac67-9ca6e8da79d7"

— title: "Computational reverse mathematics and foundational analysis"

PROJECT ARCHITECTURE

Temmie Roberts

Firestore: user management

Search by email address, phone number or user UID					Add user		
Identifier	Providers	Created ↓	Signed in	User UID			
jameshoeey@email.com		4 May 2023	4 May 2023	r9u3pit5WePxyBoWwcPRQ22GI972			
tem@tem.tem		3 Apr 2023	3 Apr 2023	N8Nqw2os3DUdPTVJI2OHNZLloir2			
test2@test.test		29 Mar 2023	3 Apr 2023	4KVL0XgtPsYF5heSkEWQtA7K7fi2			

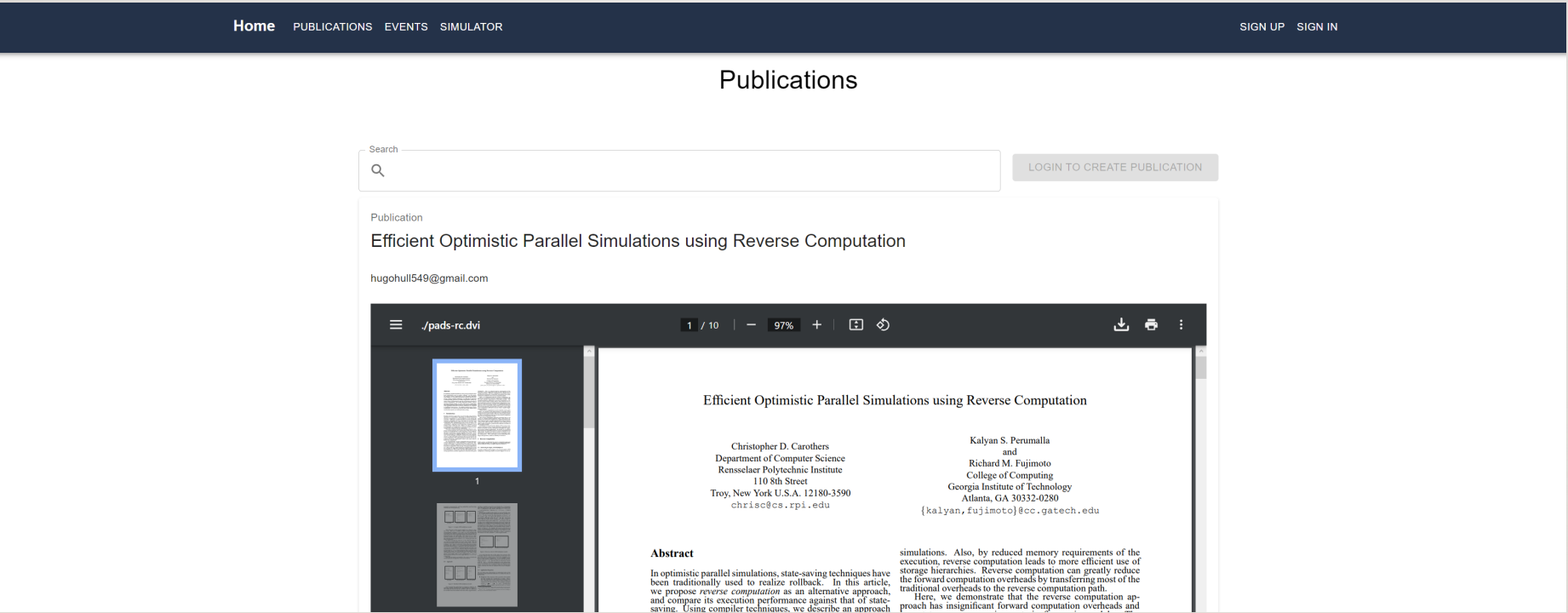
Firestore: Realtime database

```
https://group-19-17882-default-rtdb.europe-west1.firebaseio.com/app/  
└─ Events  
  └─ 0a2dcb2b-367c-4023-8cc7-1ea1dc289778  
    ├── account: "test@test.test"  
    ├── description: "Conference to discuss reverse computation"  
    ├── endTime: "2023-04-30T11:00:00.000Z"  
    ├── location: "University of Leicester"  
    ├── startTime: "2023-04-29T11:00:00.000Z"  
    └── title: "Reverse Computation Conference"  
  └─ 118731d1-7b4f-4db5-bf90-e9027ed15a30  
    ├── account: "test@test.com"  
    ├── description: "A week where there is non stop content on reverse computation"  
    ├── endTime: "2023-04-27T11:00:00.000Z"  
    ├── location: "London"  
    └── startTime: "2023-03-31T23:00:00.000Z"
```

PROJECT ARCHITECTURE

Temmie Roberts

React: Prebuilt pdf viewer component



DESIGN DECISIONS OF THE PROJECT

Teodor Sula

Simulator design choices:

- Written entirely in JavaScript, can be updated in real time
- React allowed us to use JSX files which meant we could be more flexible with our coding solutions
- The use steps and lines to differentiate between being into an if loop or not.

DESIGN DECISIONS OF THE PROJECT

Teodor Sula

Code running:

```
x=5
if (false){
x=x+1
}elseif (true){
x=x*3
}else{
x=x*x
}
y=((12*12)-4)/10
y=y/10
z=x+1
```

Current step: 1 Current line: 1

STEP BACKSTEP FORWARD

Memory:

- Name: Value:
- Name: x Value: 5

Reverse Computation Simulator

Code running:

```
x=5
if (false){
x=x+1
}elseif (true){
x=x*3
}else{
x=x*x
}
y=((12*12)-4)/10
y=y/10
z=x+1
```

Current step: 2 Current line: 3

STEP BACKSTEP FORWARD

Memory:

- Name: Value:
- Name: x Value: 5

FUTURE PLANS

Lee Milligan

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- Expanding the simulator
 - Iteration
 - Functions
 - Expanding moderation
 - User connectivity